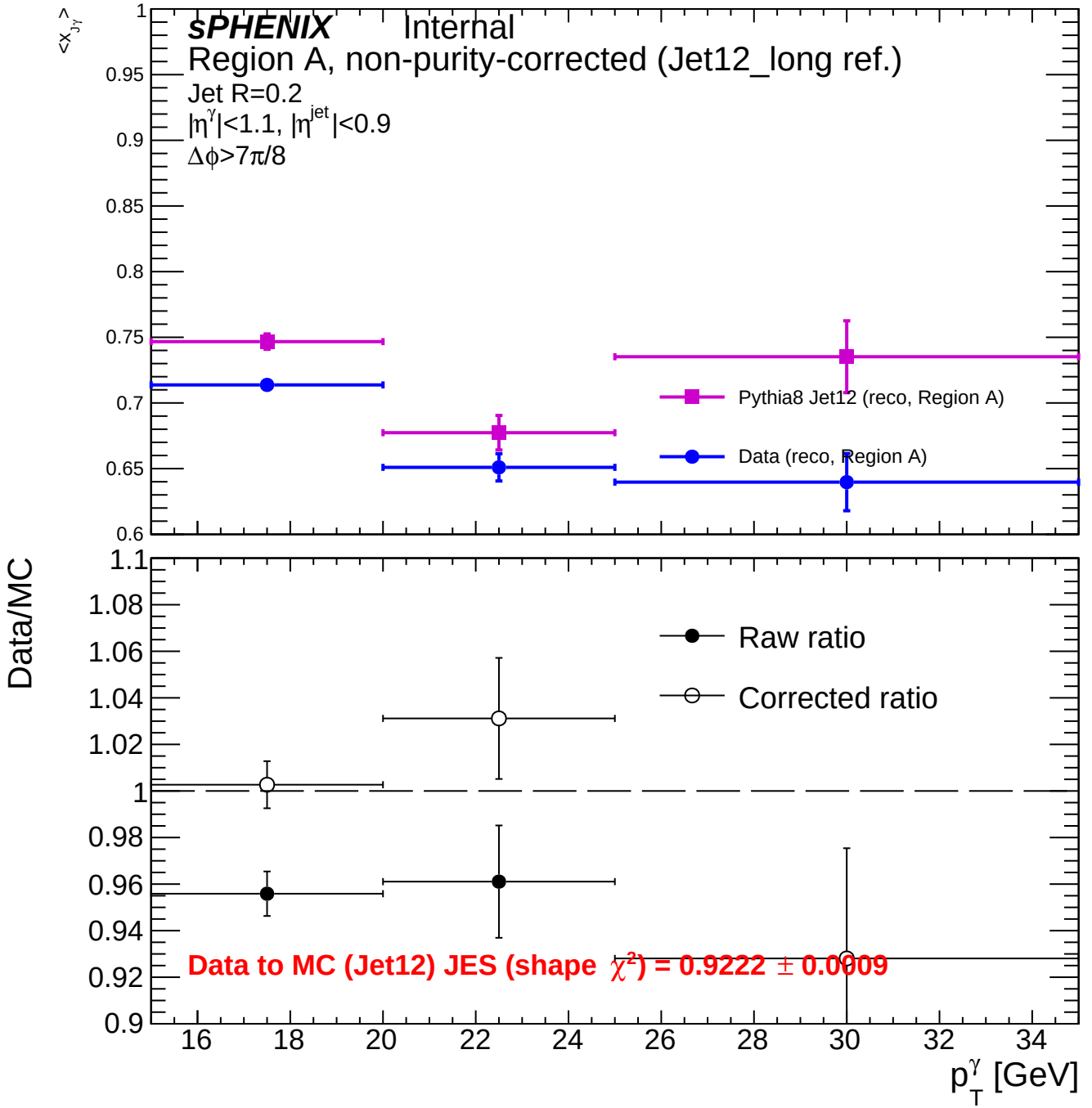
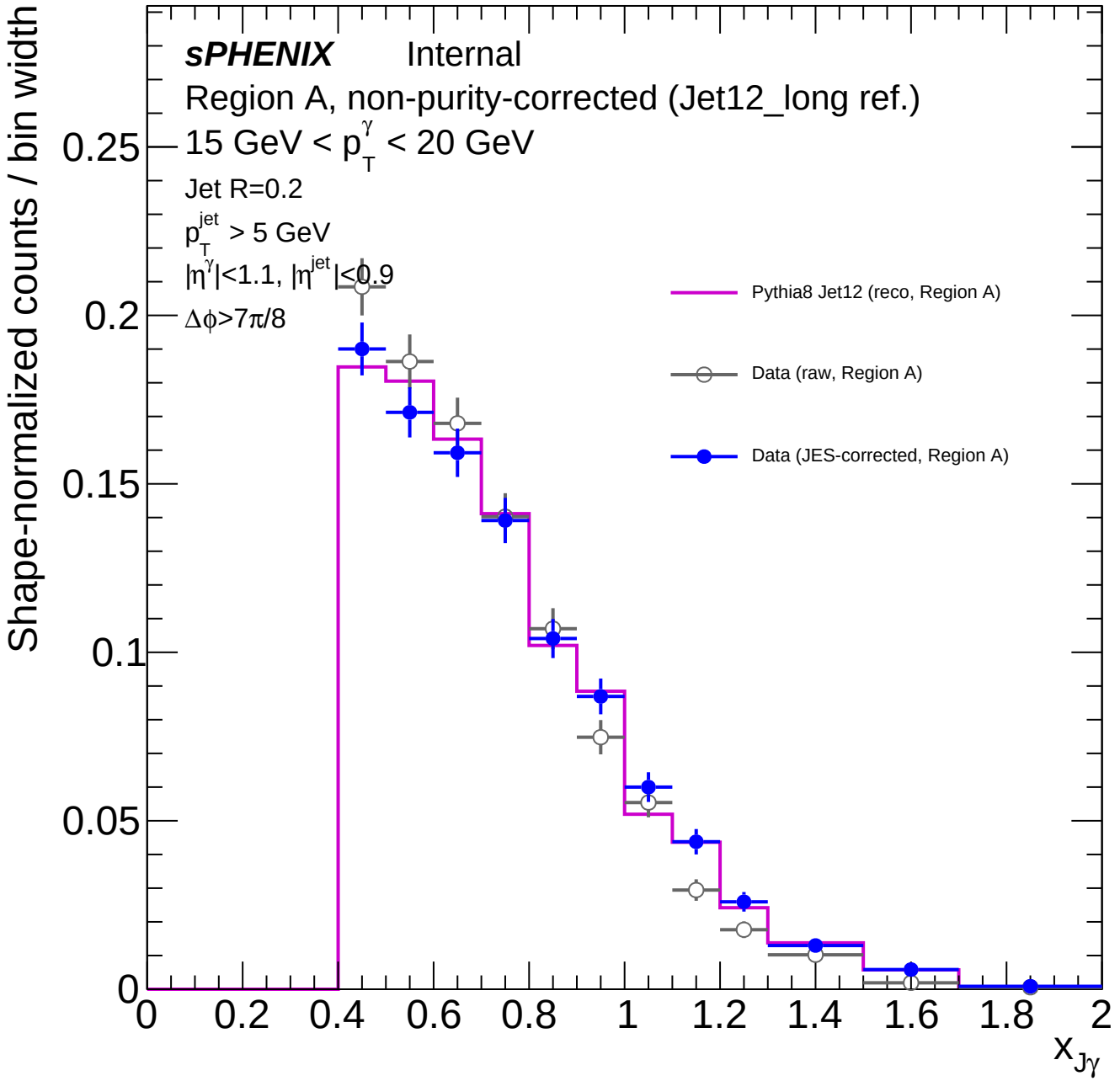
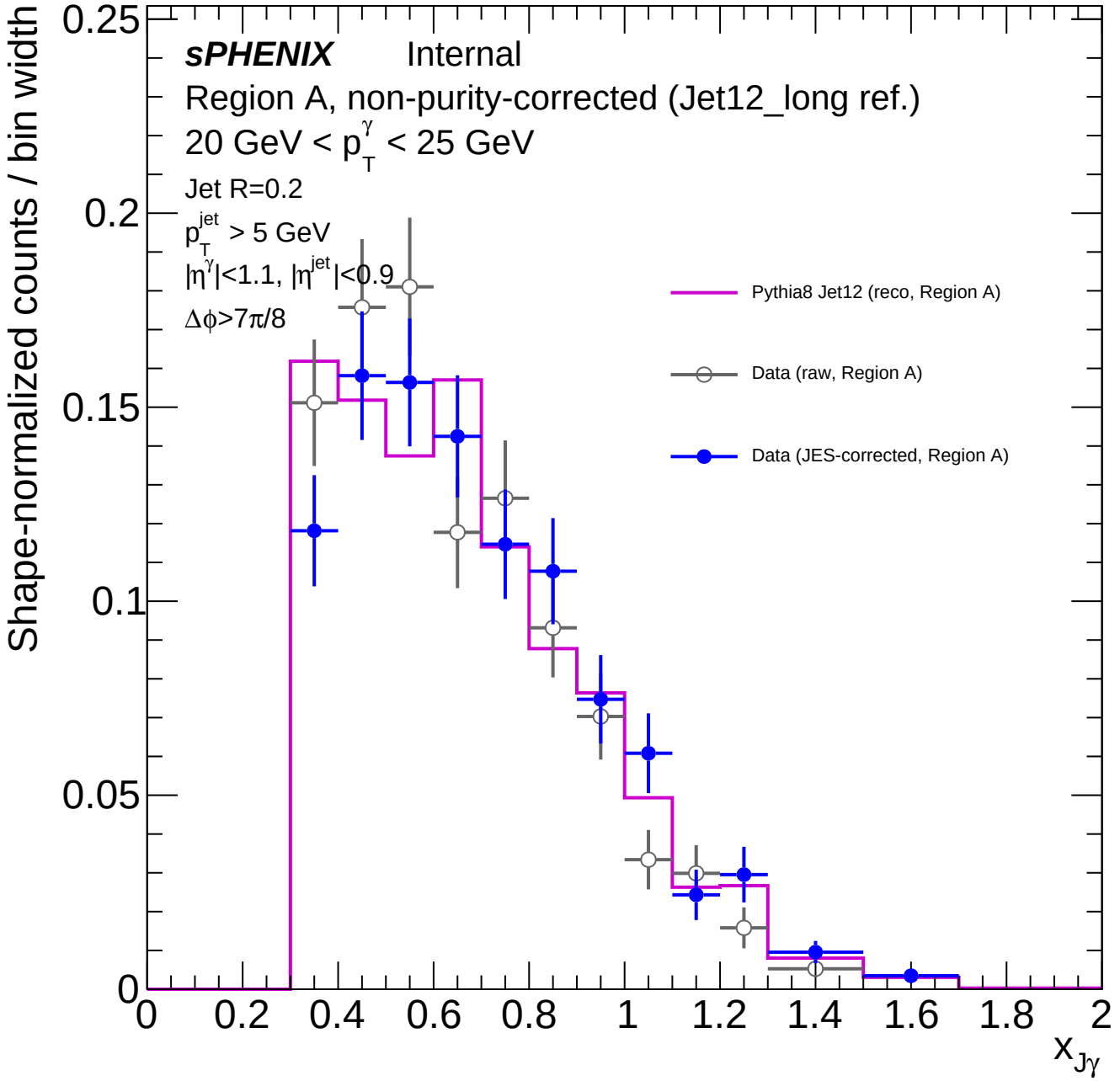
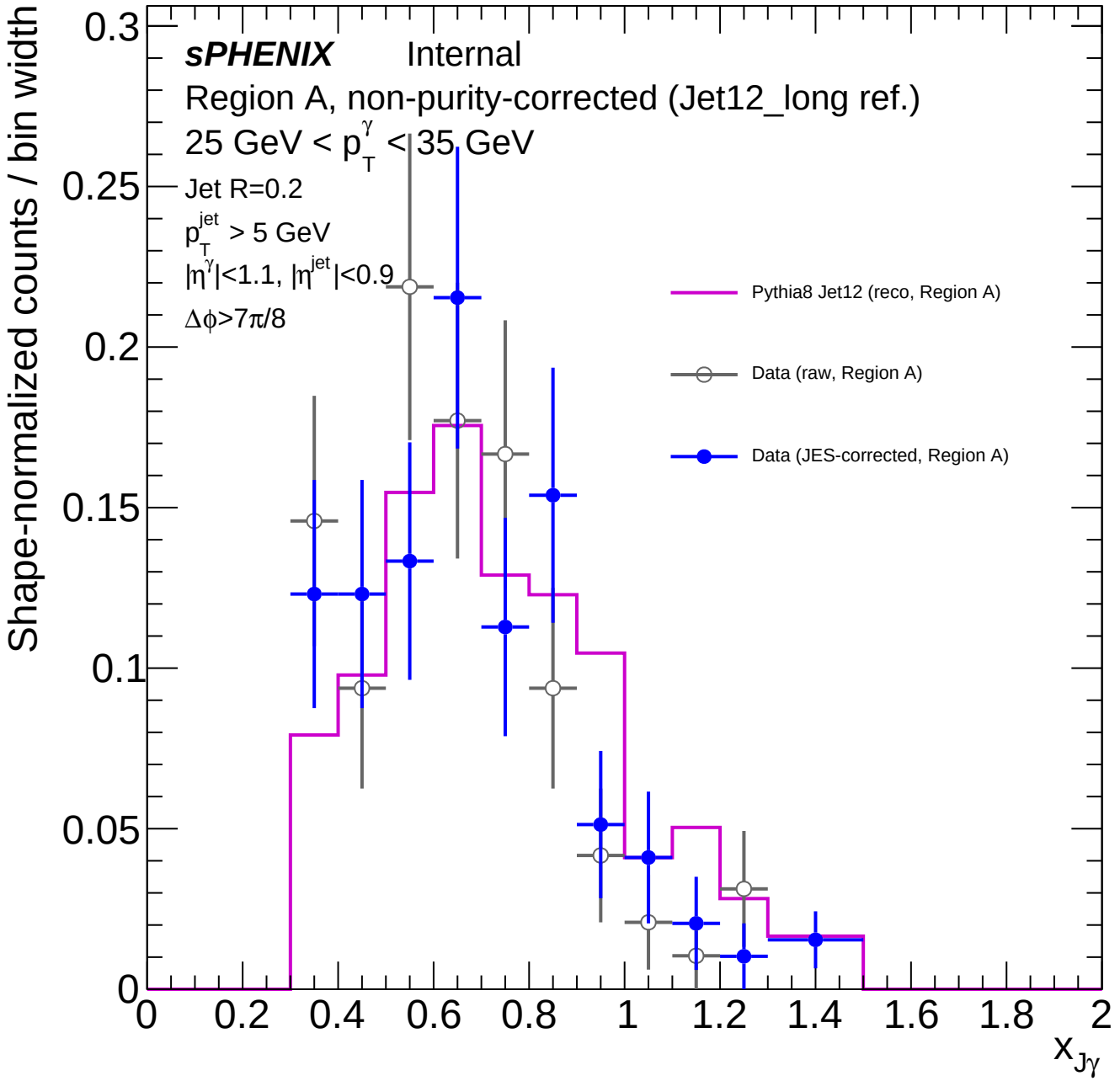
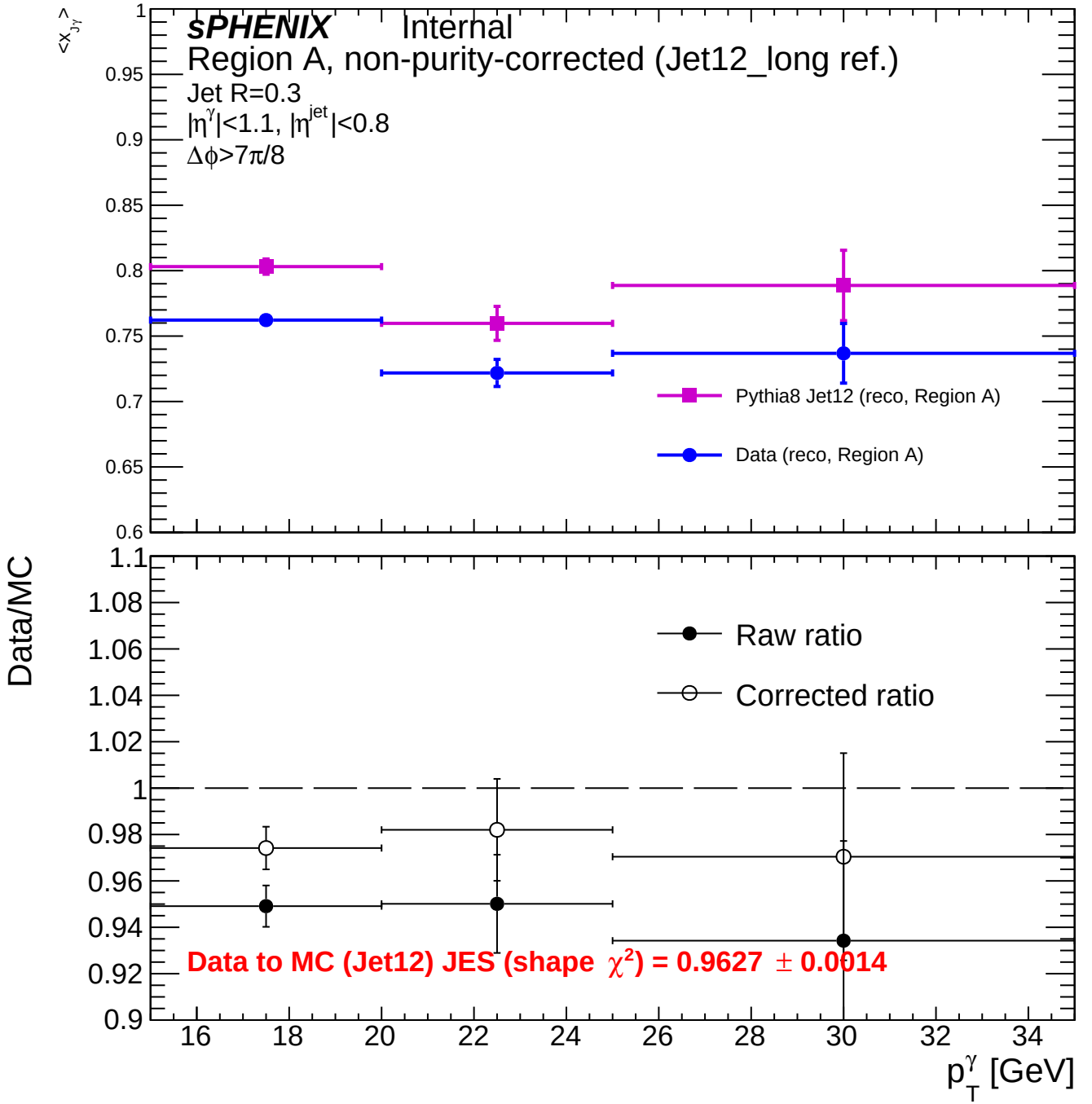
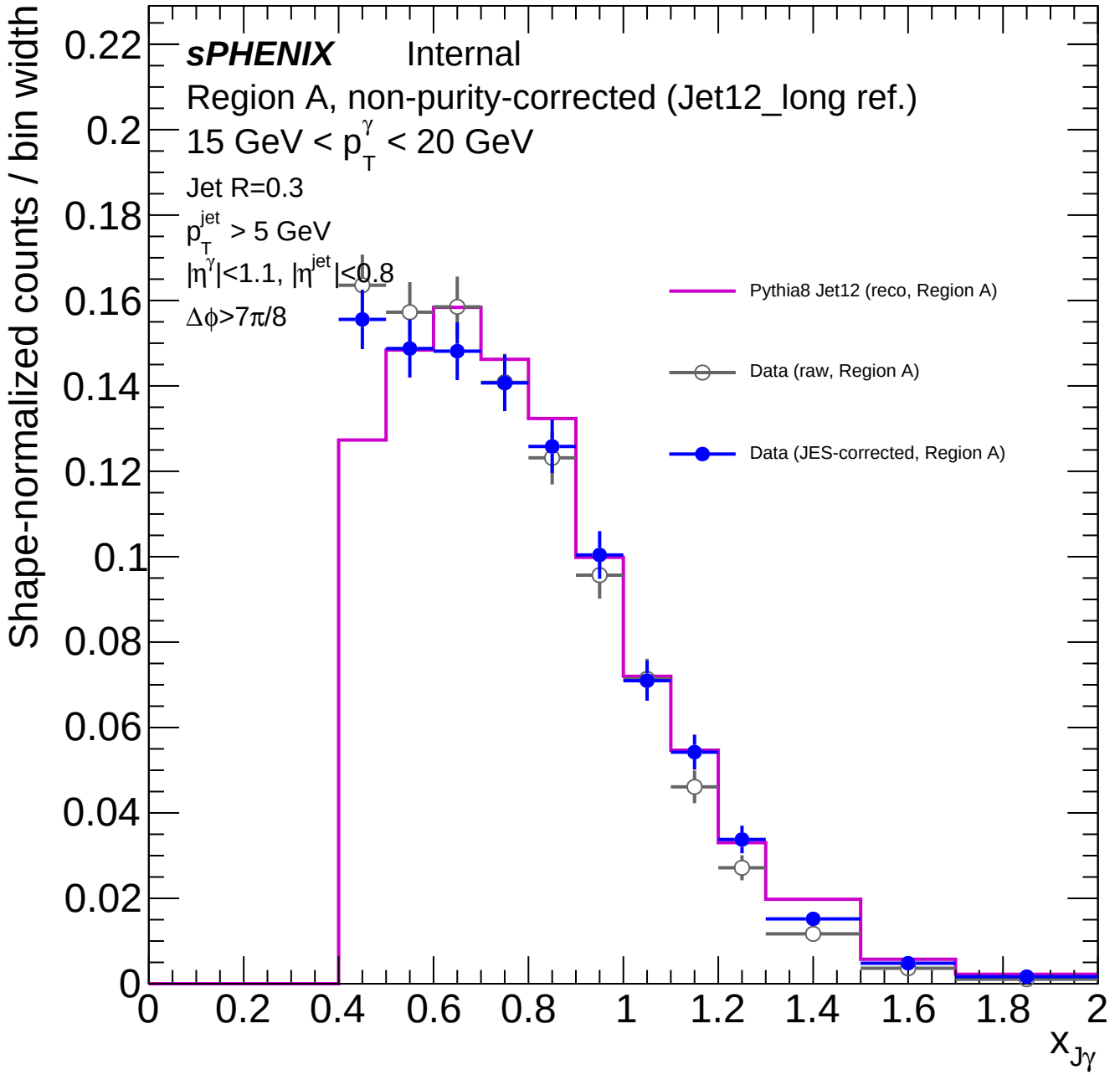

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
Jet R=0.2
|η^γ|<1.1, |η^jet|<0.9
Δφ>7π/8
<x_Jγ>
Pythia8 Jet12 (reco, Region A)
Data (reco, Region A)
Data/MC
Raw ratio
Corrected ratio
Data to MC (Jet12) JES (shape χ²) = 0.9222 ± 0.0009
p_T^γ [GeV]

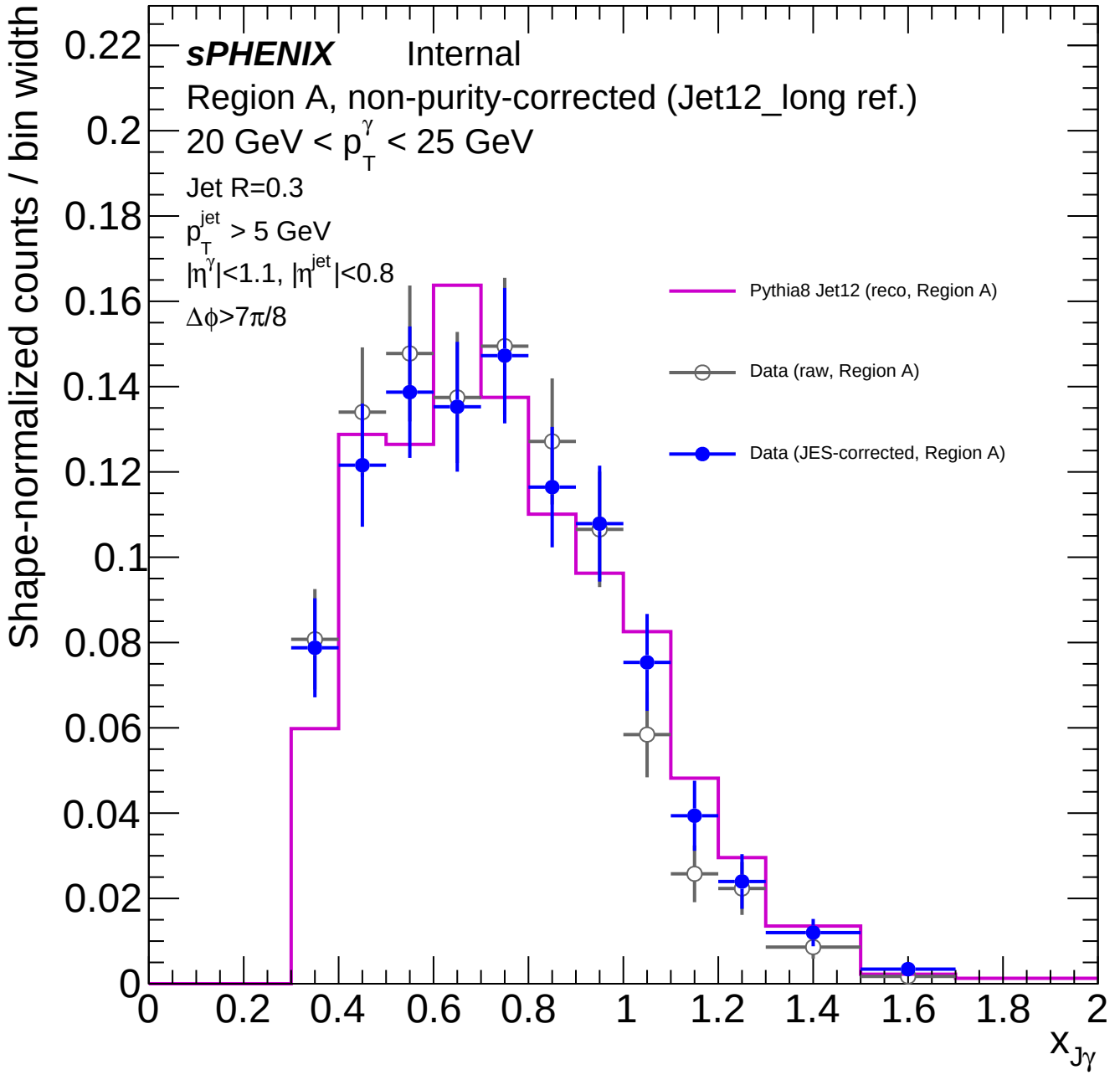

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.2
p_T^{jet} > 5 GeV
$|\eta^\gamma|<1.1$, $|\eta^{jet}|<0.9$
$\Delta\phi>7\pi/8$
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
$x_{J\gamma}$
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

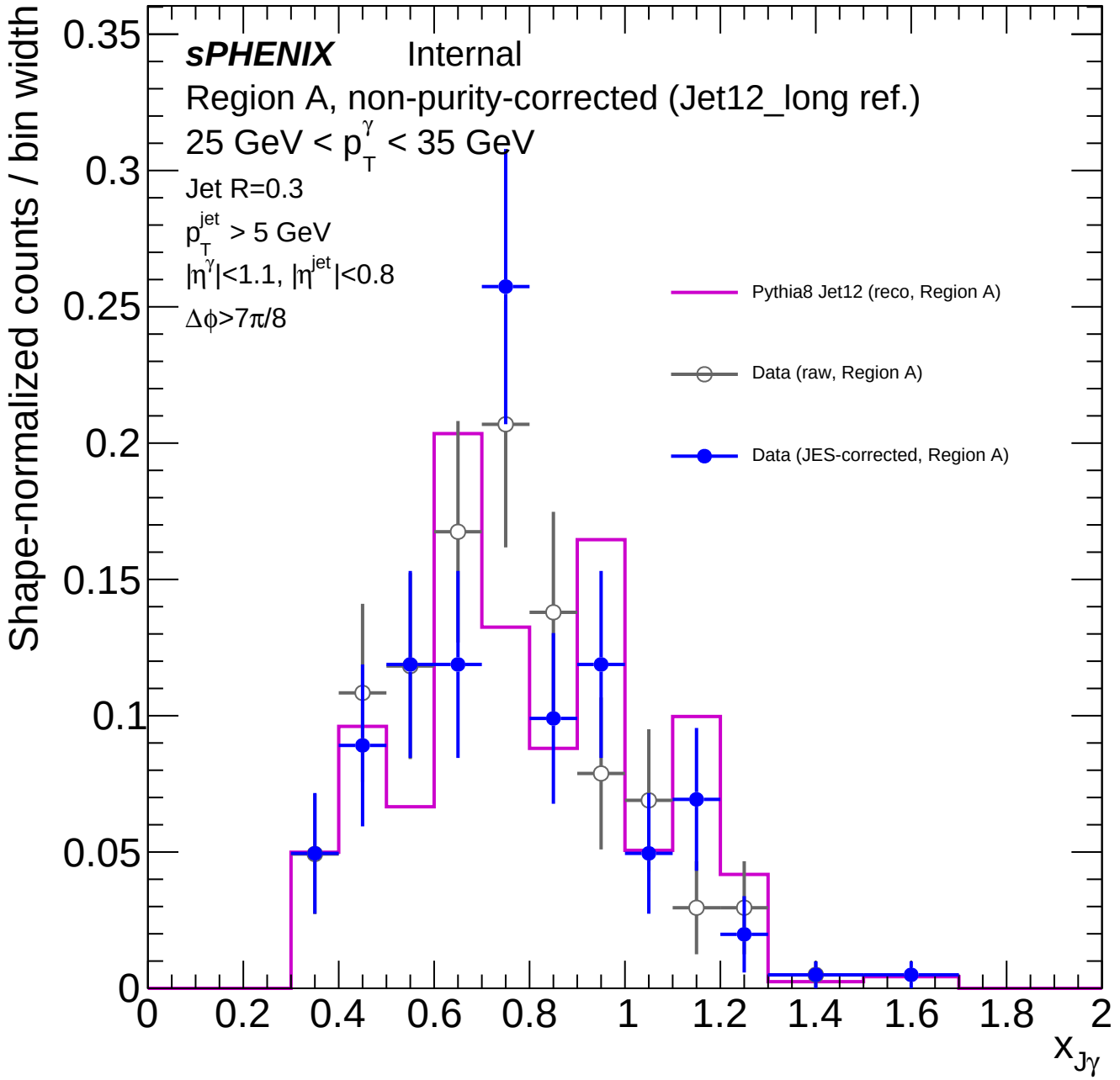

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.2
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.9
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

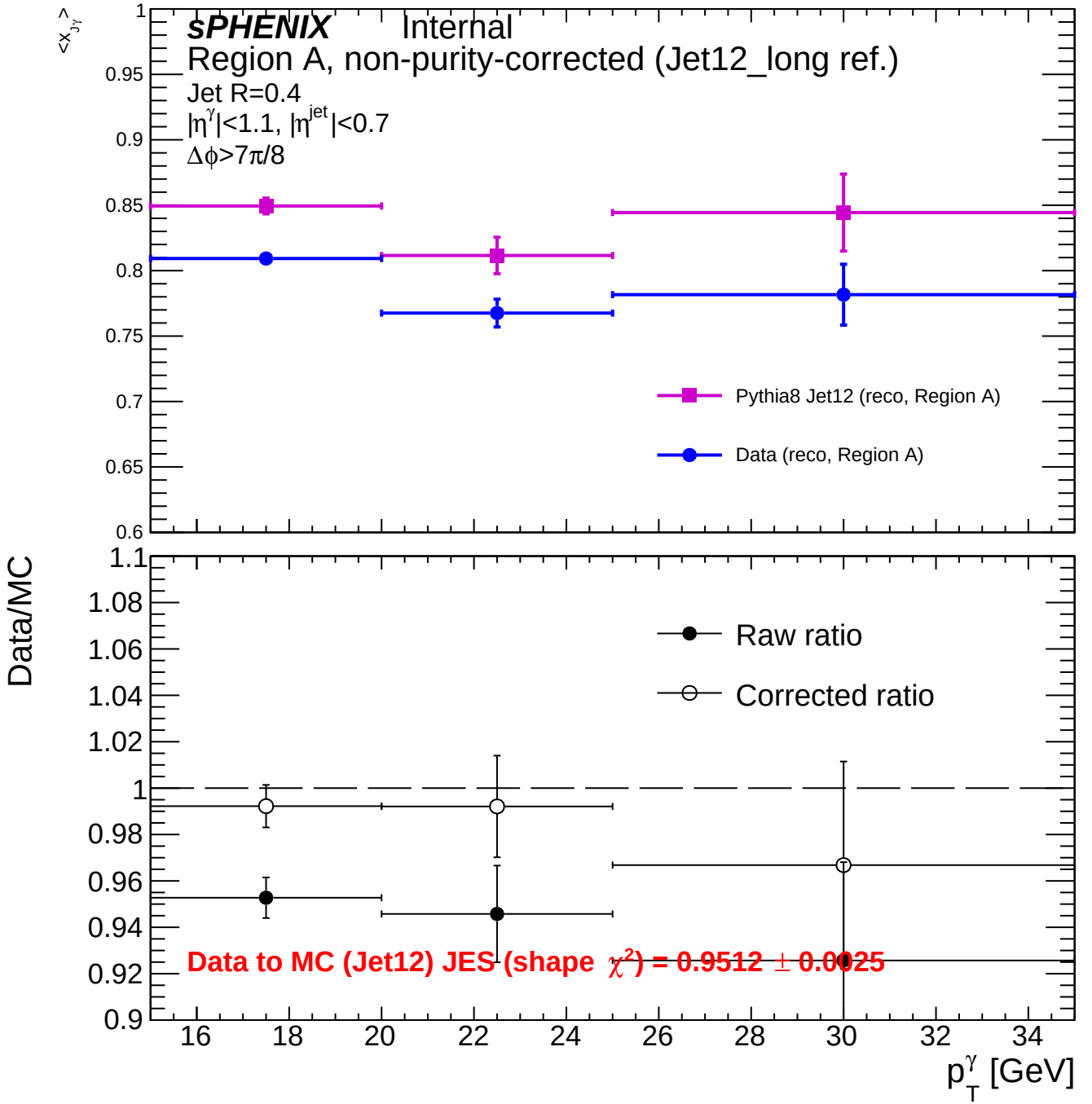

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.2
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.9
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

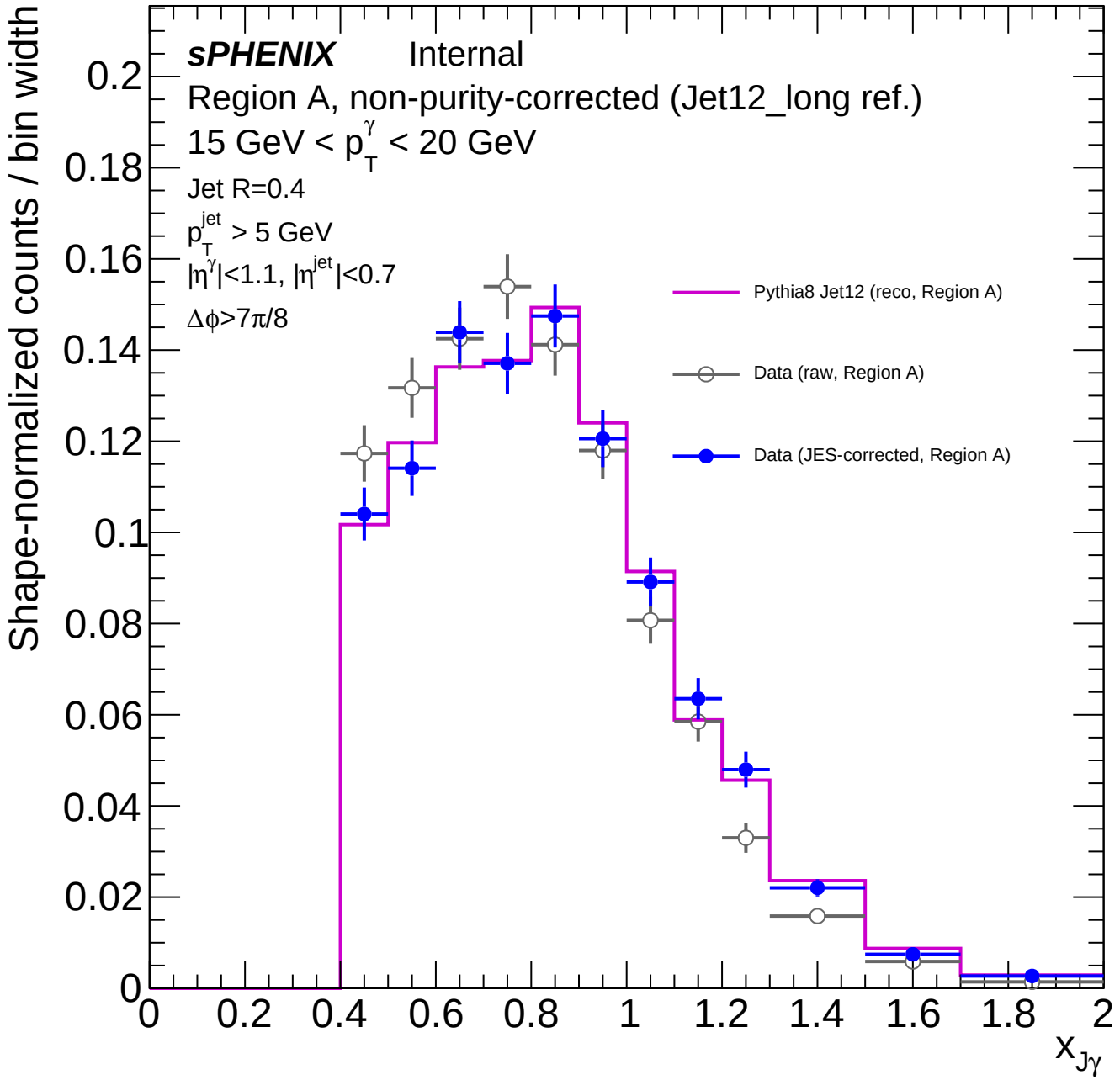

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
Jet R=0.3
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
<x_Jγ>
Pythia8 Jet12 (reco, Region A)
Data (reco, Region A)
Data/MC
Raw ratio
Corrected ratio
Data to MC (Jet12) JES (shape χ²) = 0.9627 ± 0.0014
p_T^γ [GeV]

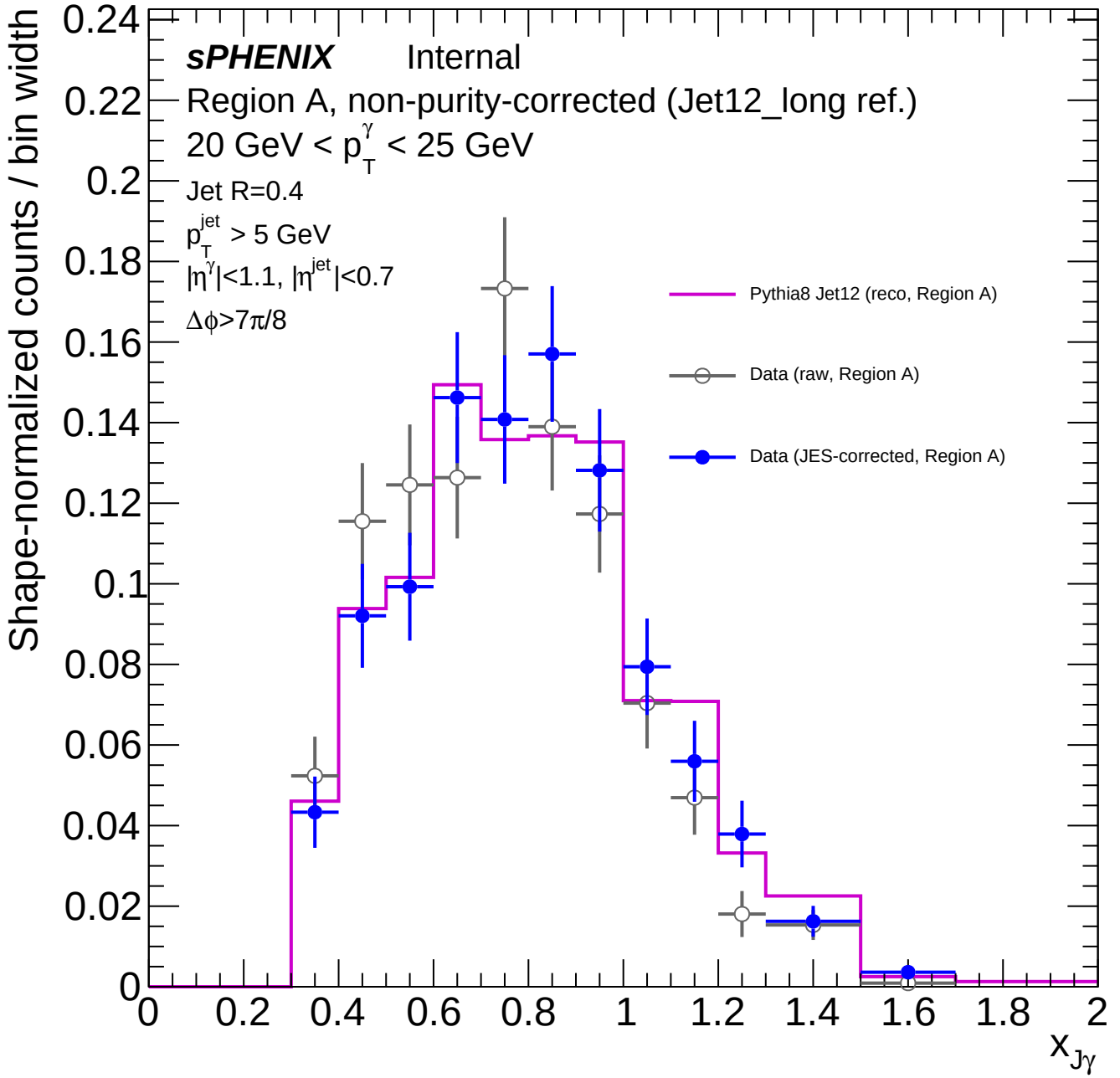

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.3
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

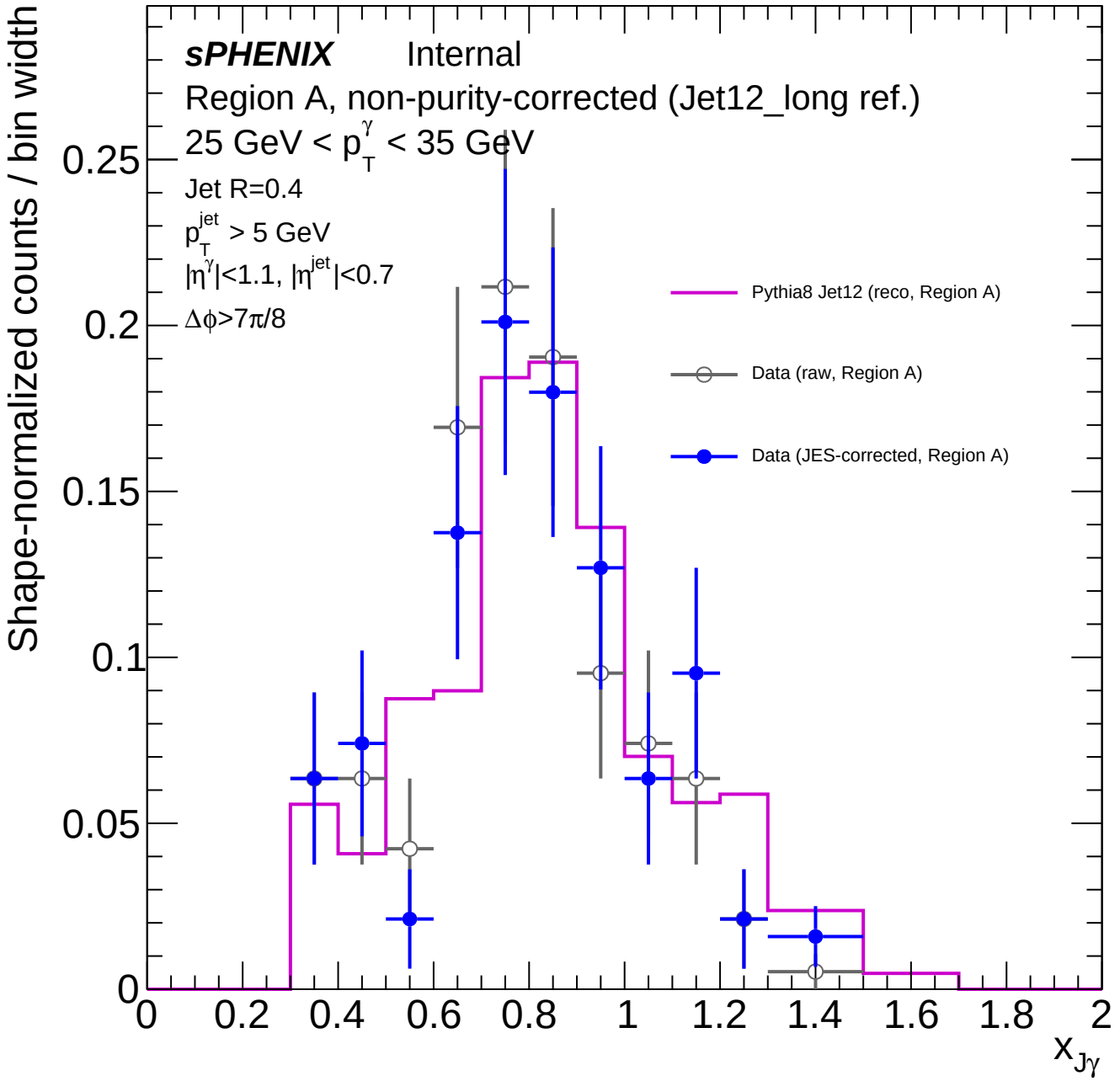

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.3
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

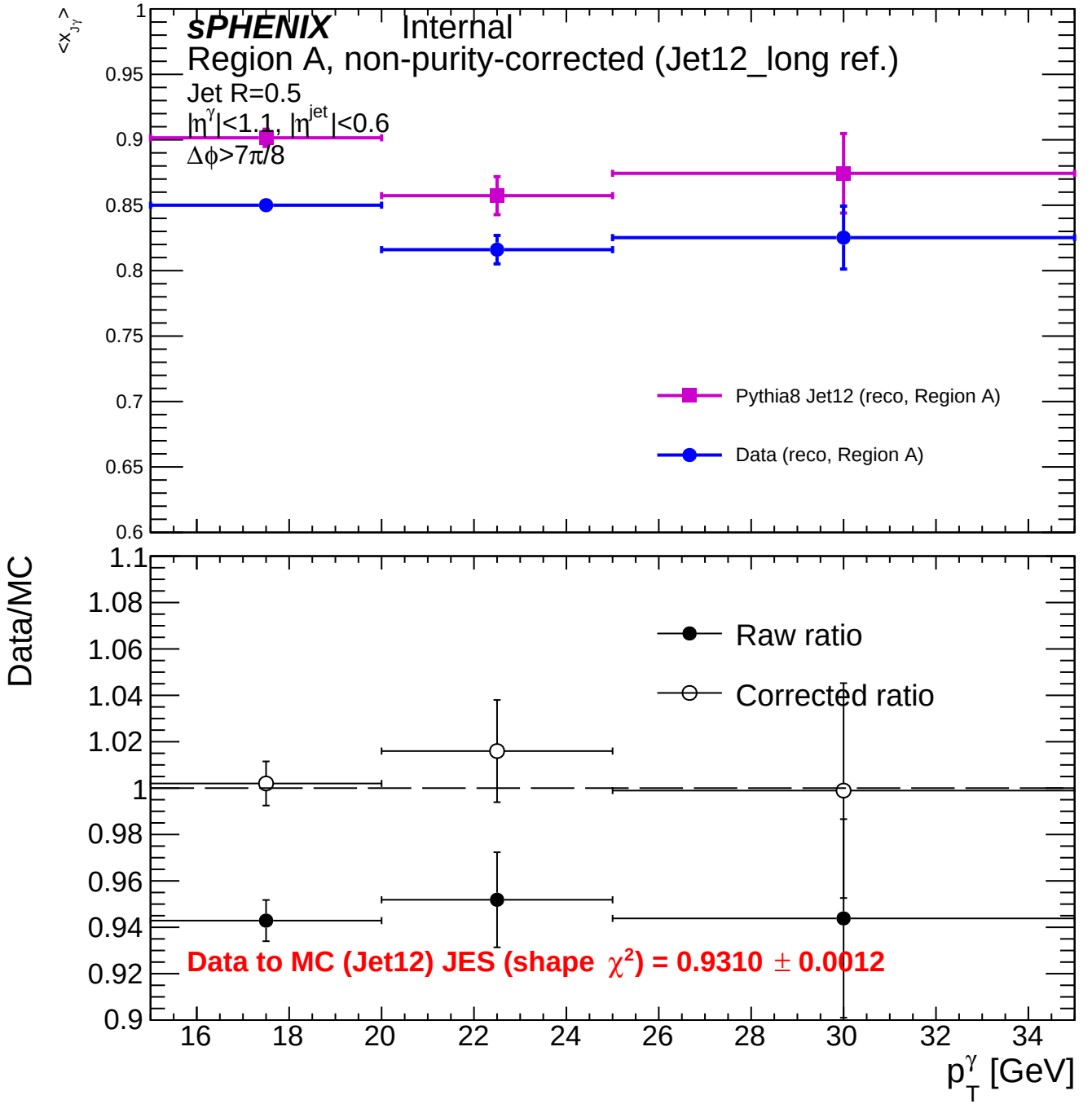

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.3
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.8
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

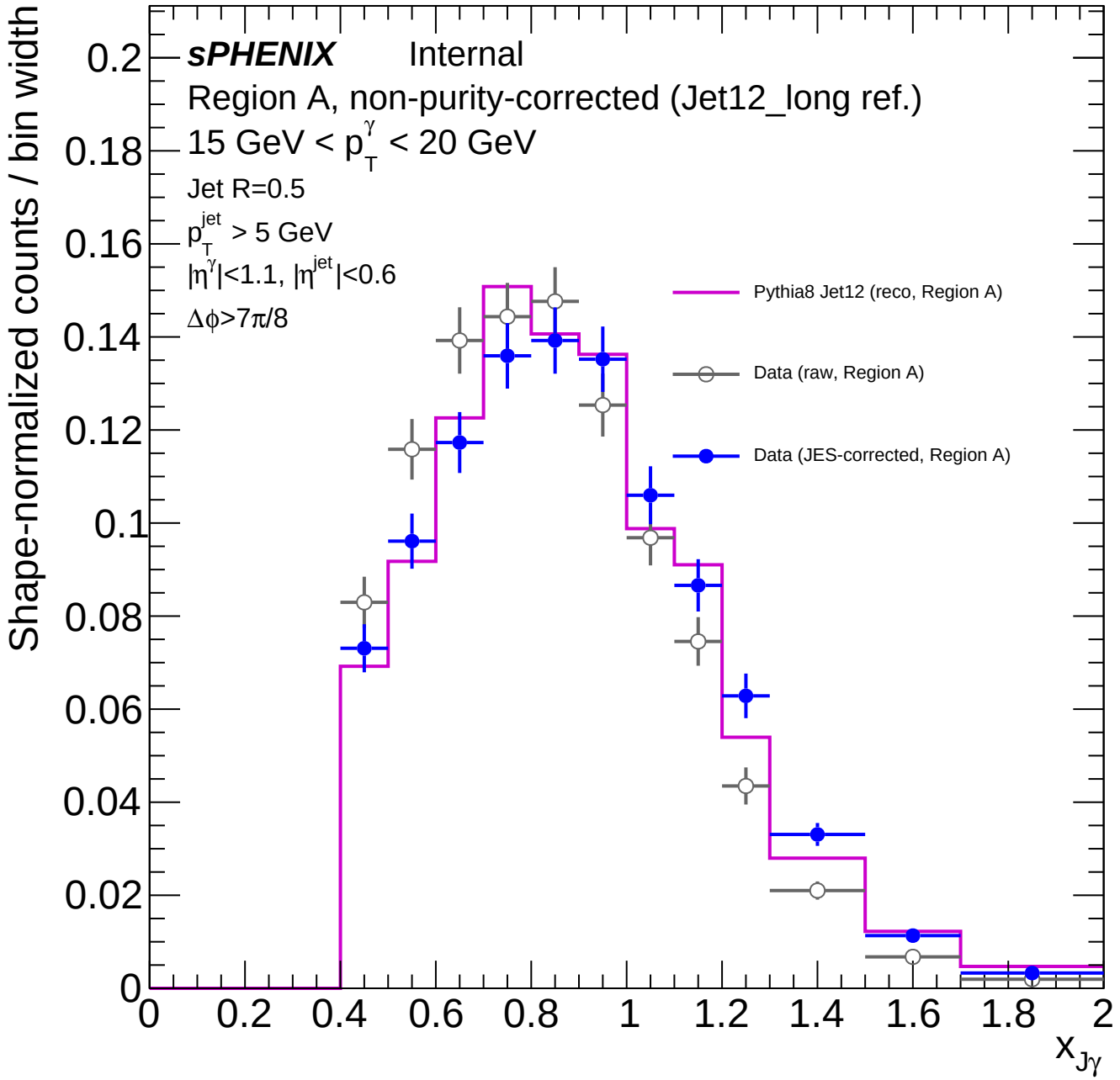

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
Jet R=0.4
|η^γ|<1.1, |η^jet|<0.7
Δϕ>7π/8
<x_Jγ>
Pythia8 Jet12 (reco, Region A)
Data (reco, Region A)
Data/MC
Raw ratio
Corrected ratio
Data to MC (Jet12) JES (shape χ²) = 0.9512 ± 0.0025
p_T^γ [GeV]

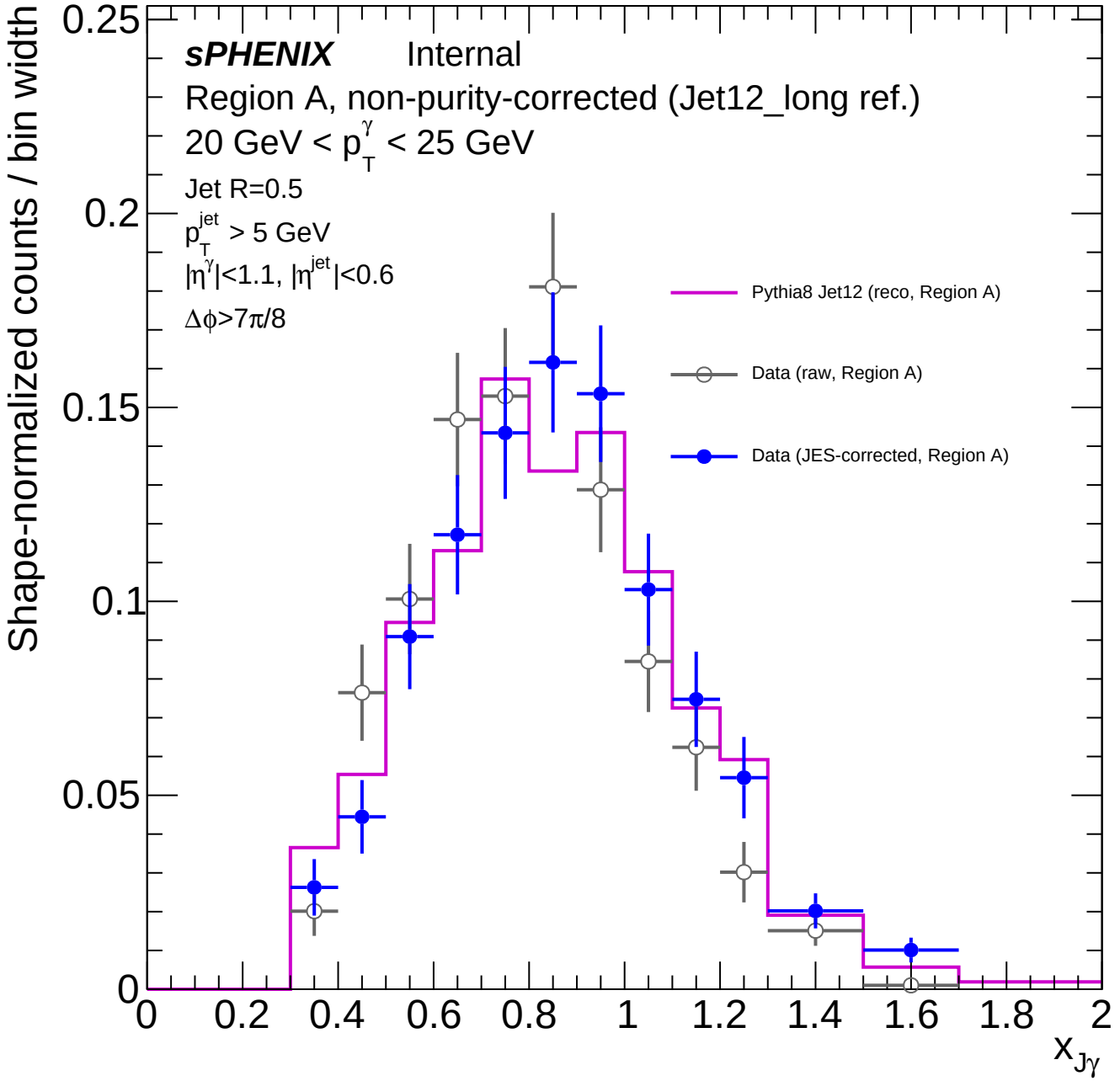

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.4
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.7
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

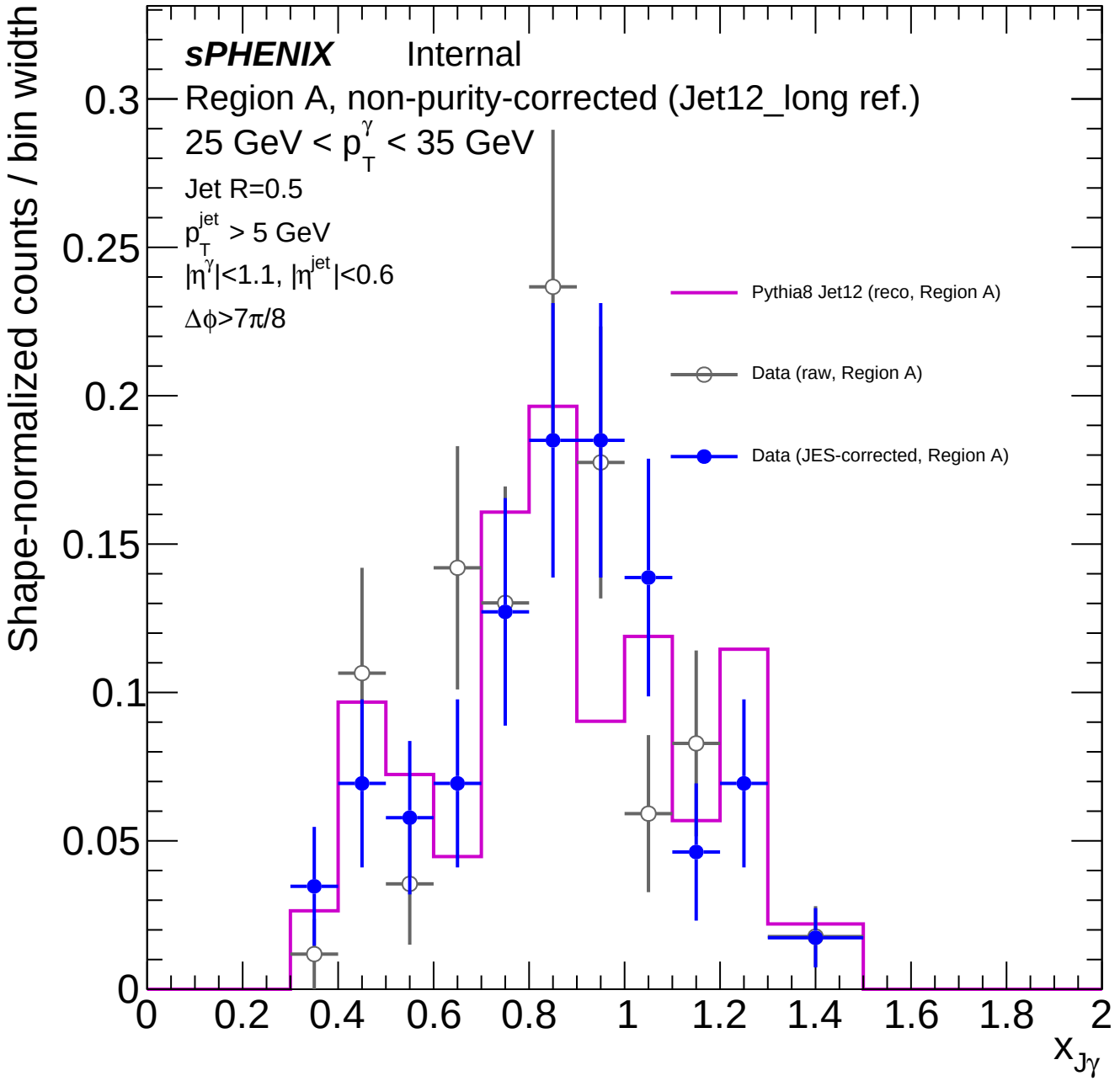

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.4
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.7
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.24
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

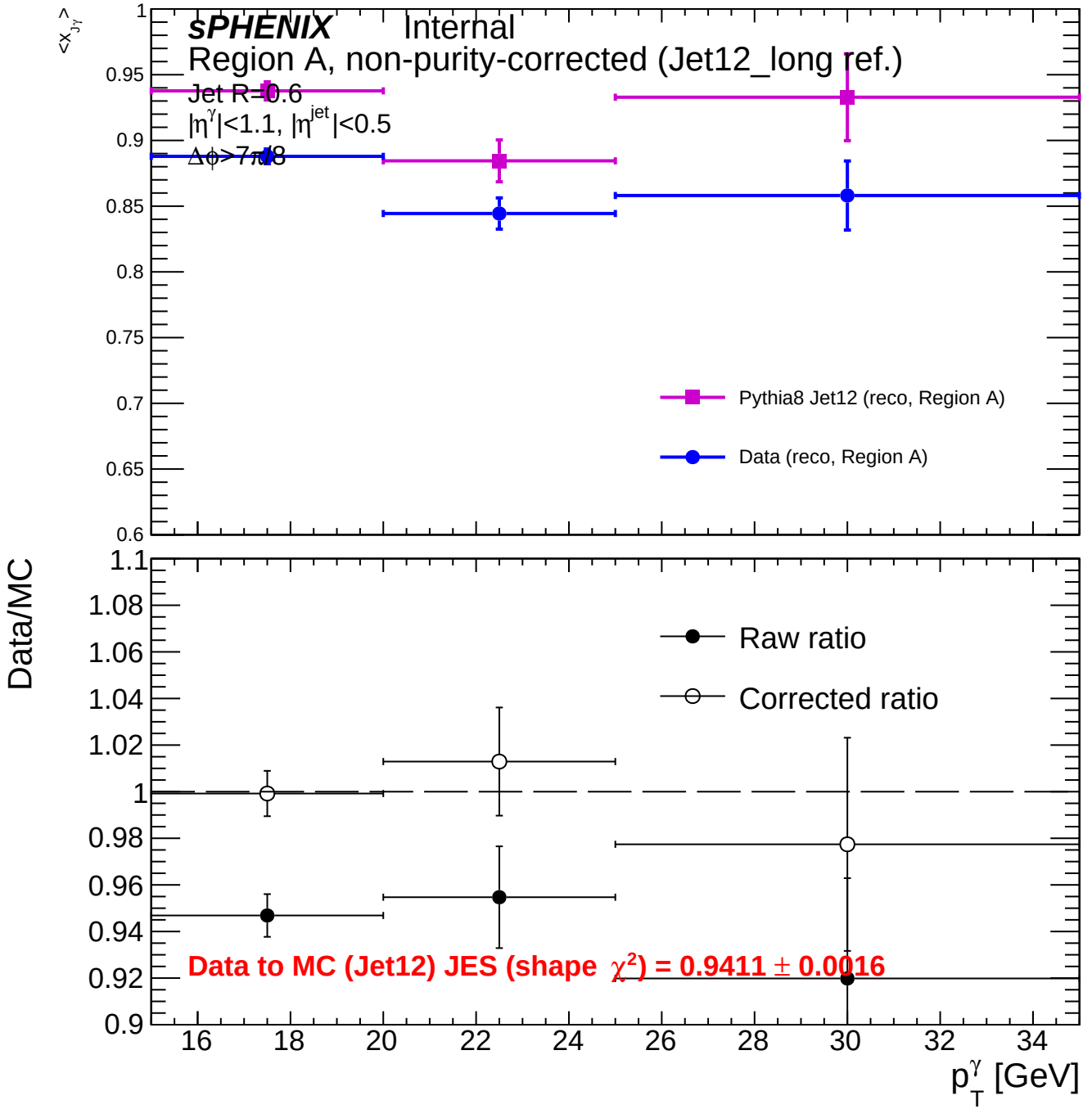

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.4
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.7
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
0
0.05
0.1
0.15
0.2
0.25
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

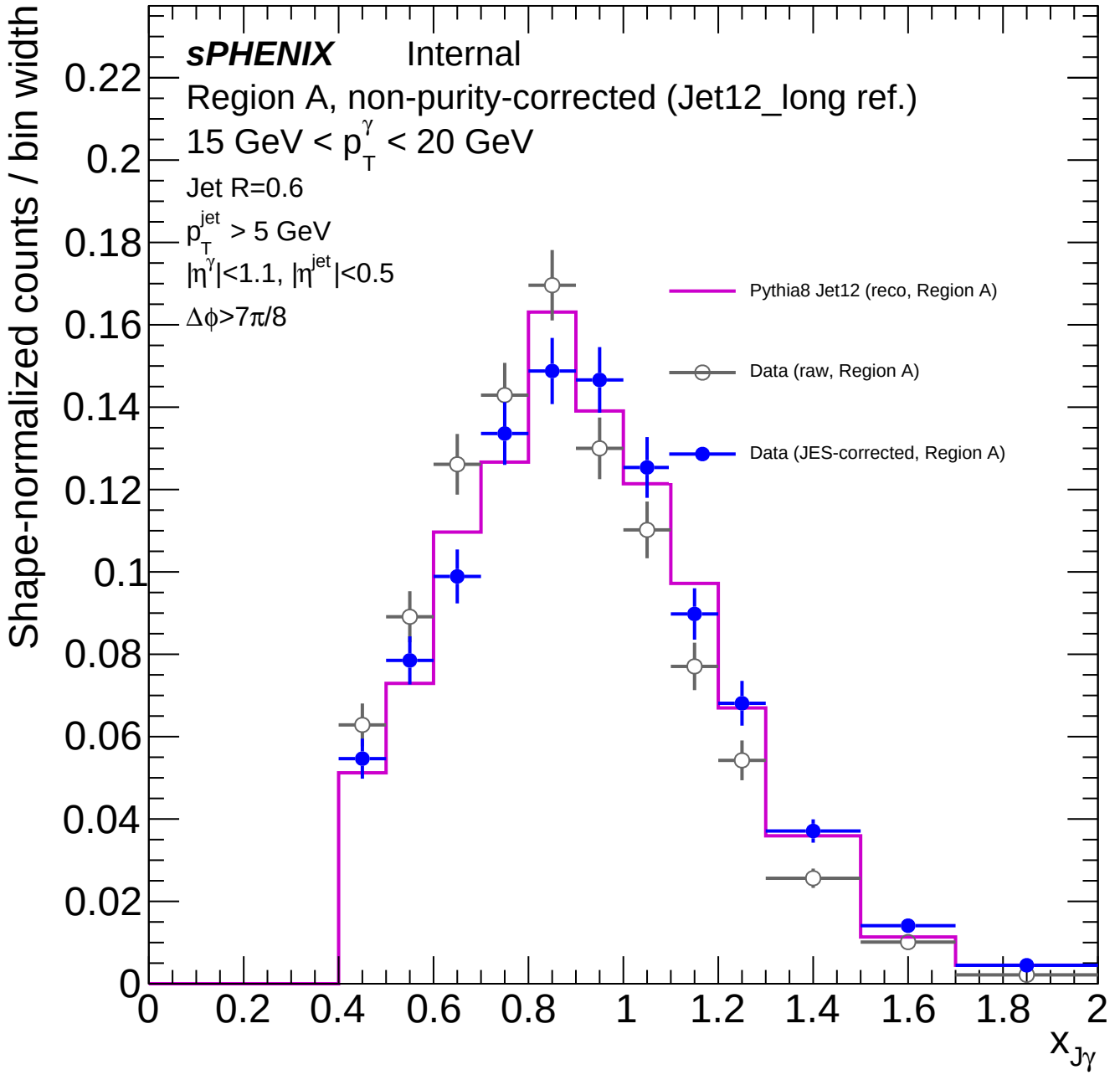

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
Jet R=0.5
|η^γ|<1.1, |η^jet|<0.6
Δϕ>7π/8
<x_jγ>
Pythia8 Jet12 (reco, Region A)
Data (reco, Region A)
Data/MC
Raw ratio
Corrected ratio
Data to MC (Jet12) JES (shape χ²) = 0.9310 ± 0.0012
p_T^γ [GeV]

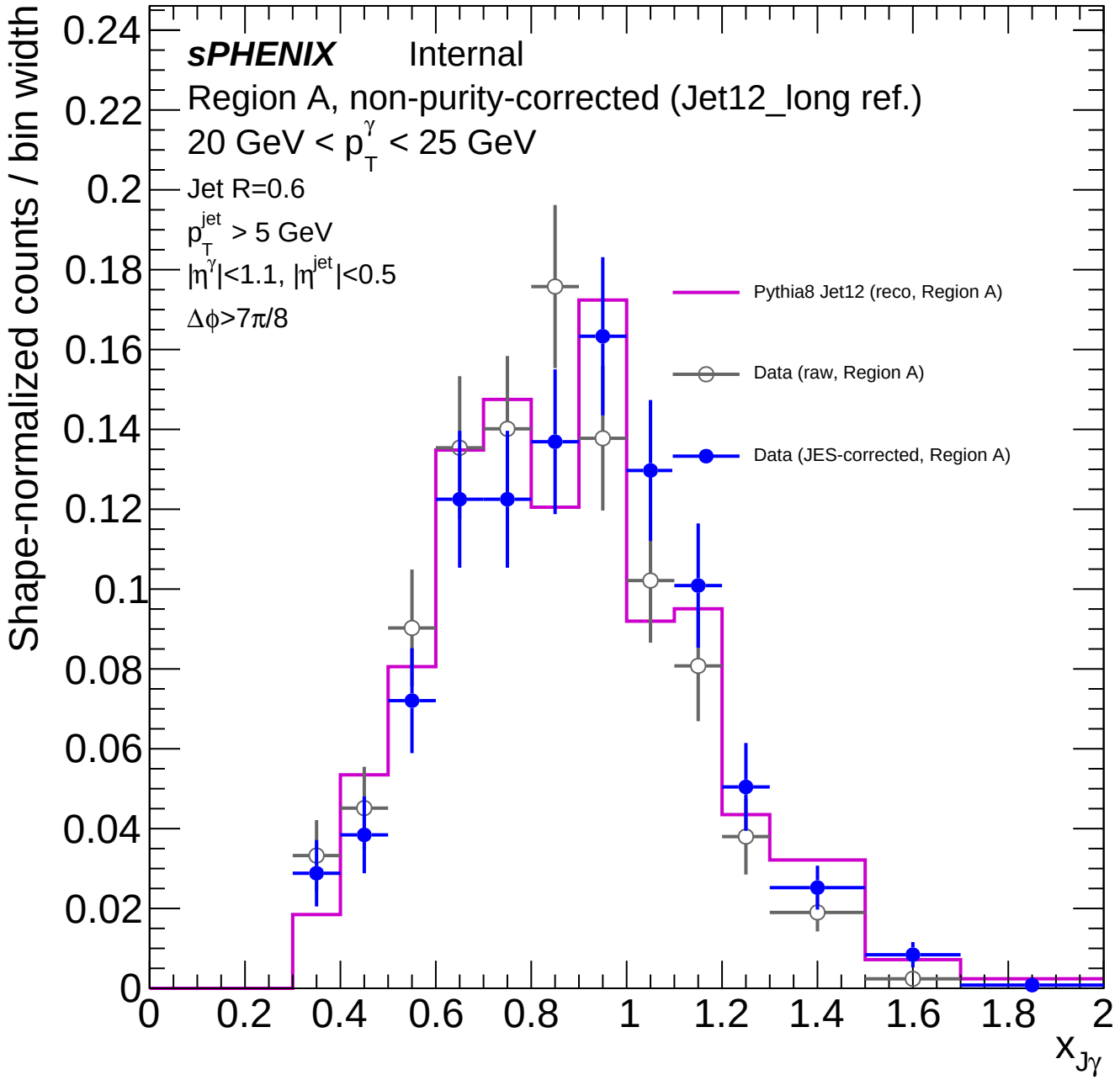

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.5
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.6
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

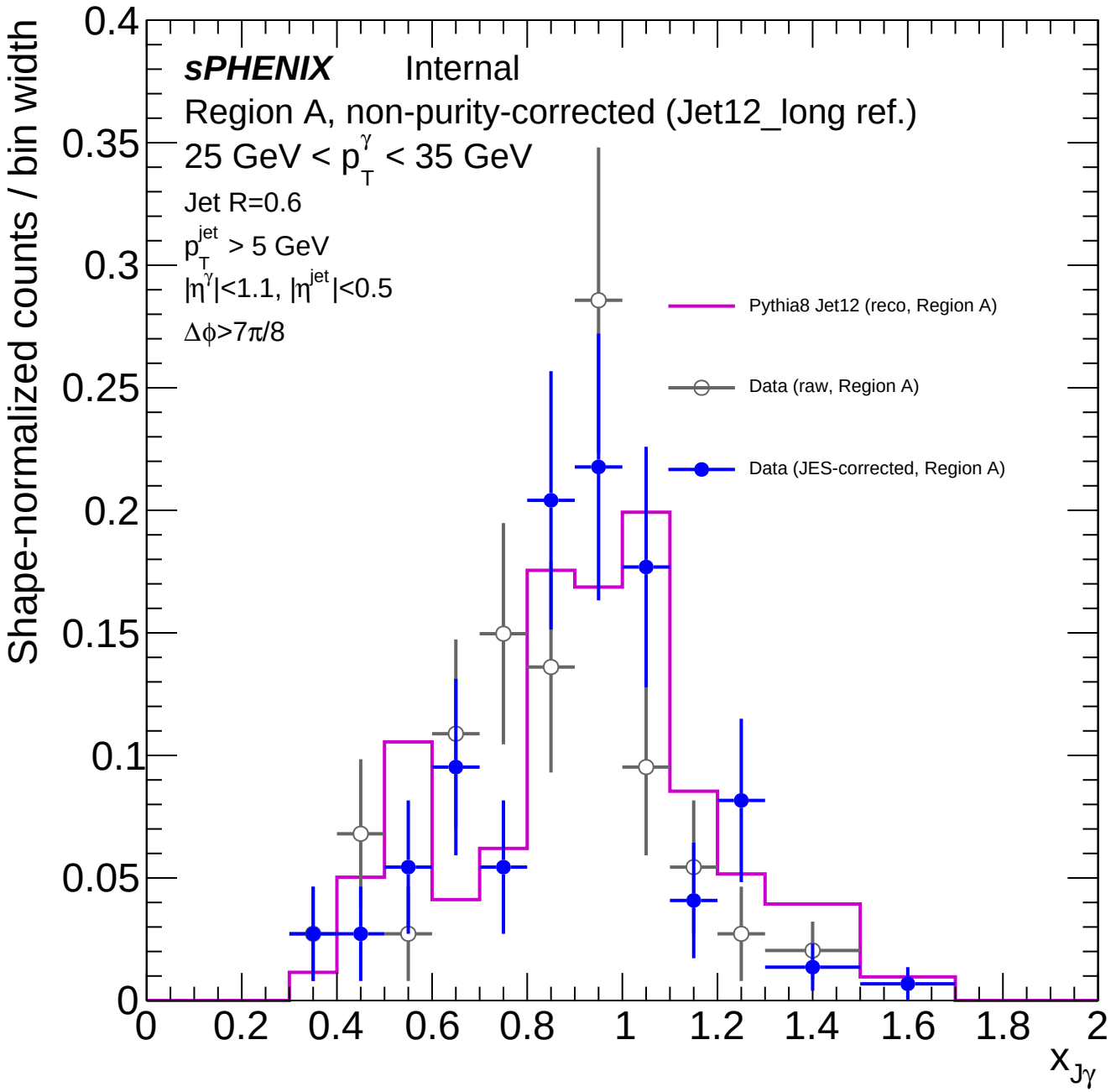

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.5
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.6
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

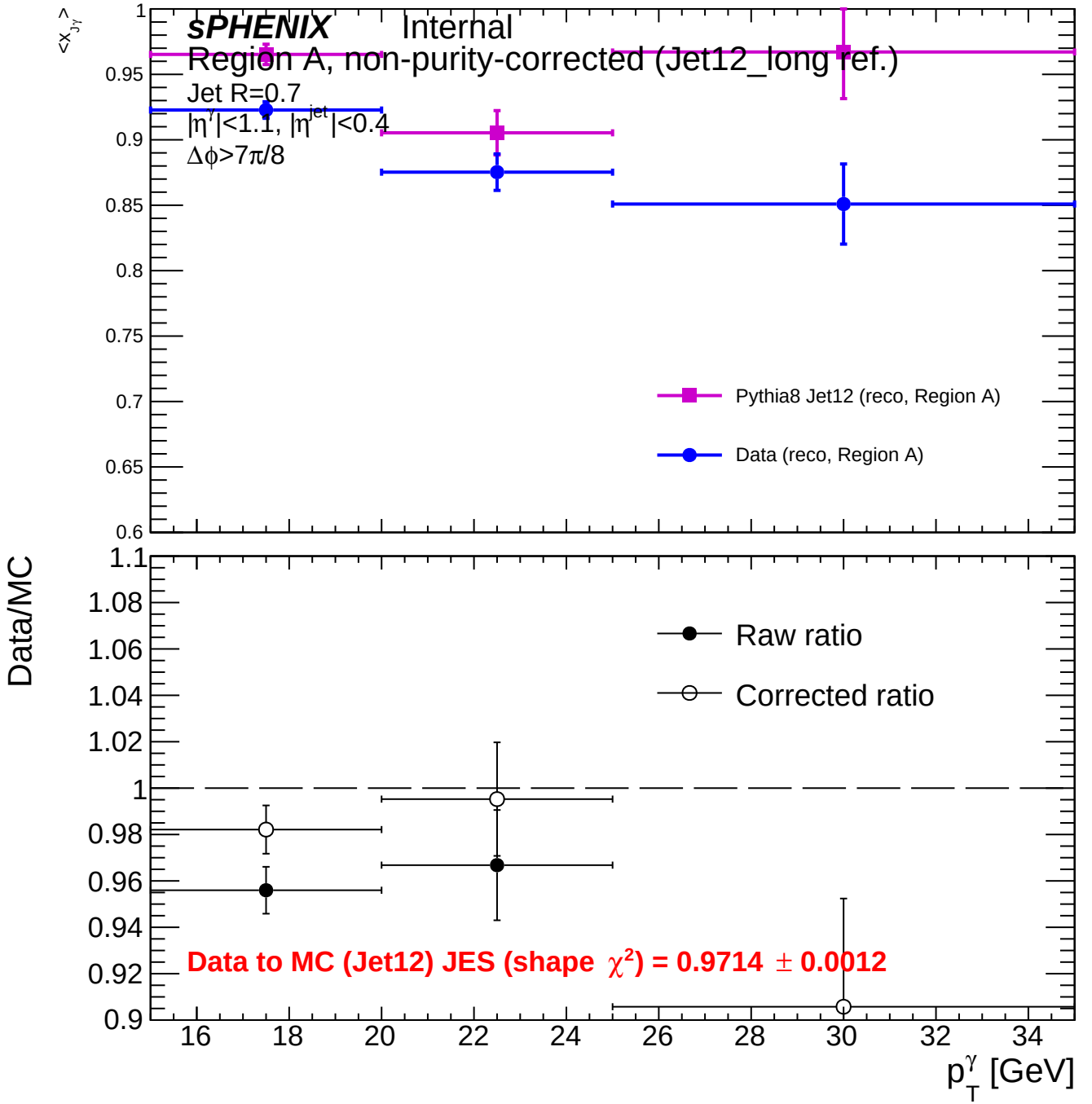

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.5
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.6
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

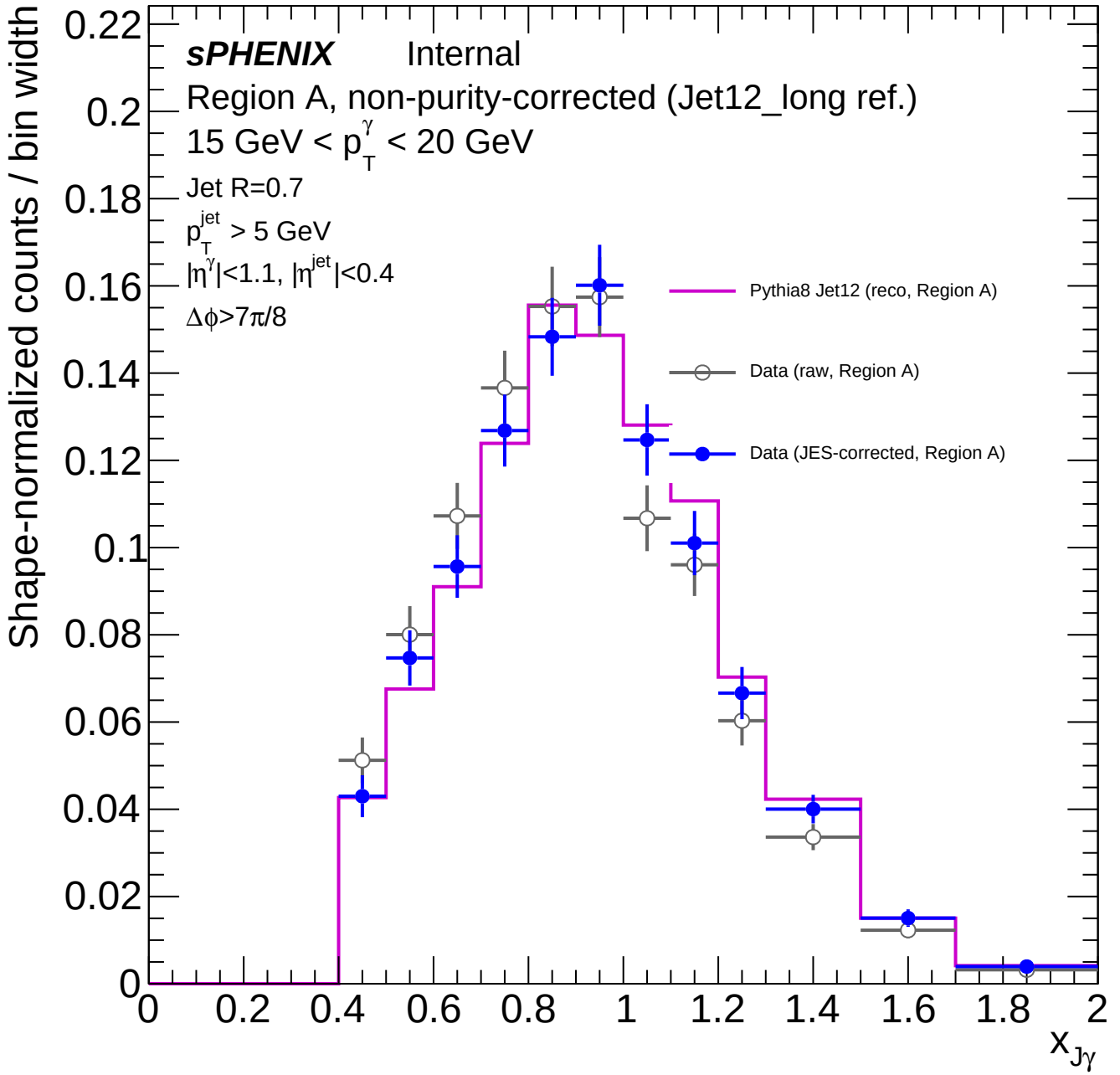

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
Jet R=0.6
|η^γ|<1.1, |η^jet|<0.5
Δφ>7π/8
<x_Jγ>
Pythia8 Jet12 (reco, Region A)
Data (reco, Region A)
Data/MC
Raw ratio
Corrected ratio
Data to MC (Jet12) JES (shape χ²) = 0.9411 ± 0.0016
p_T^γ [GeV]

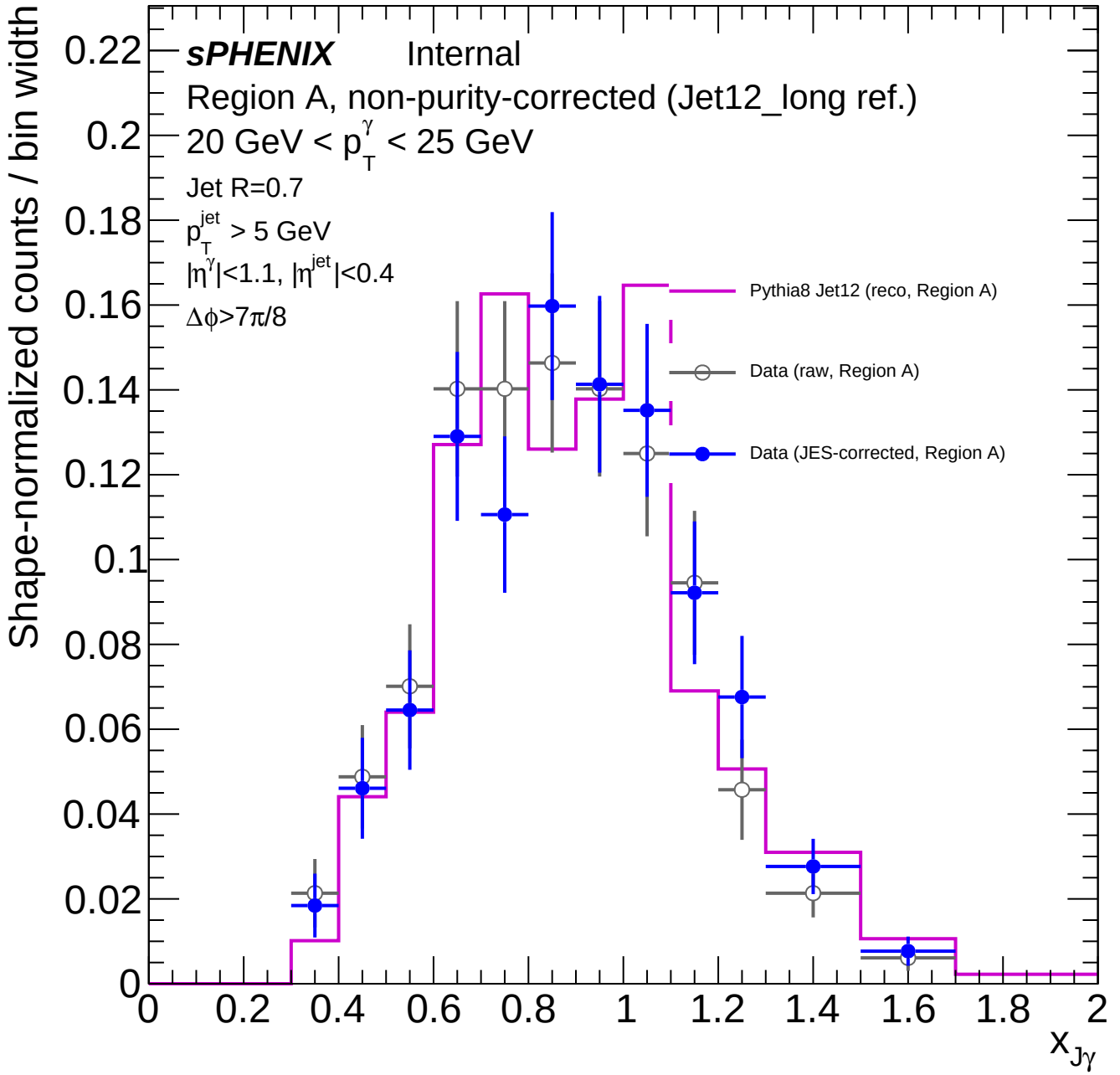

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.6
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.5
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

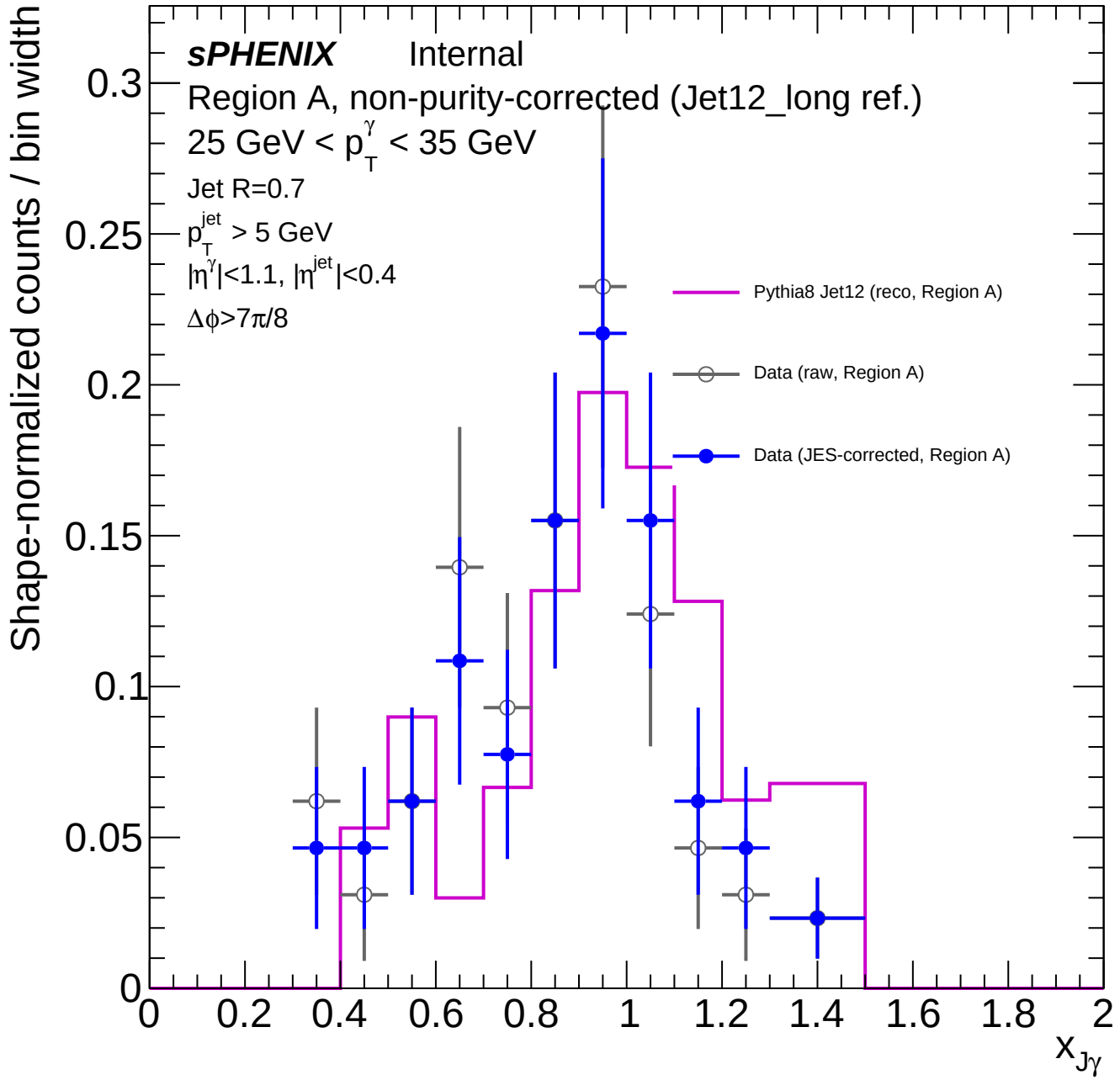

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.6
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.5
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.24
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

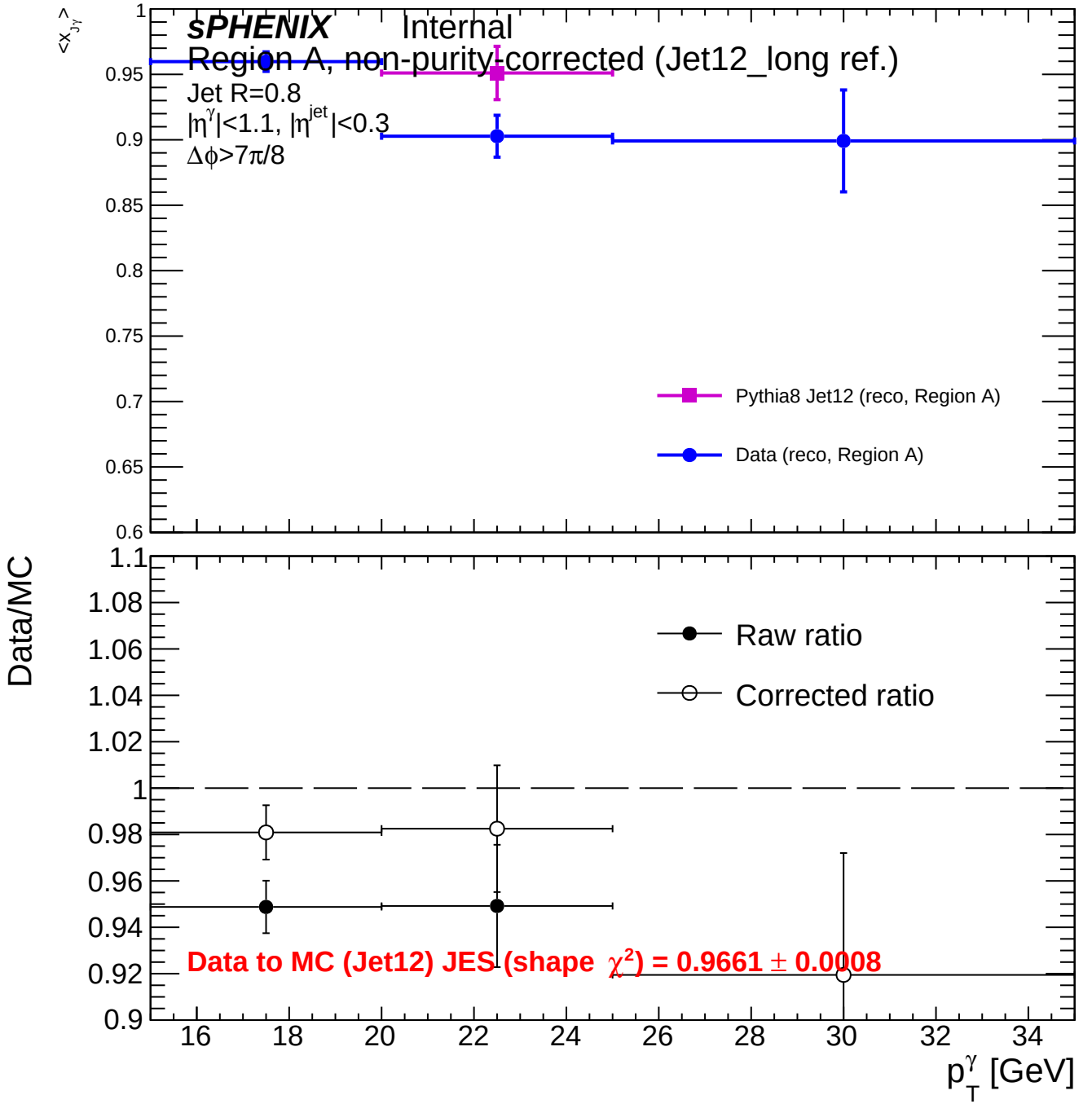

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.6
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.5
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

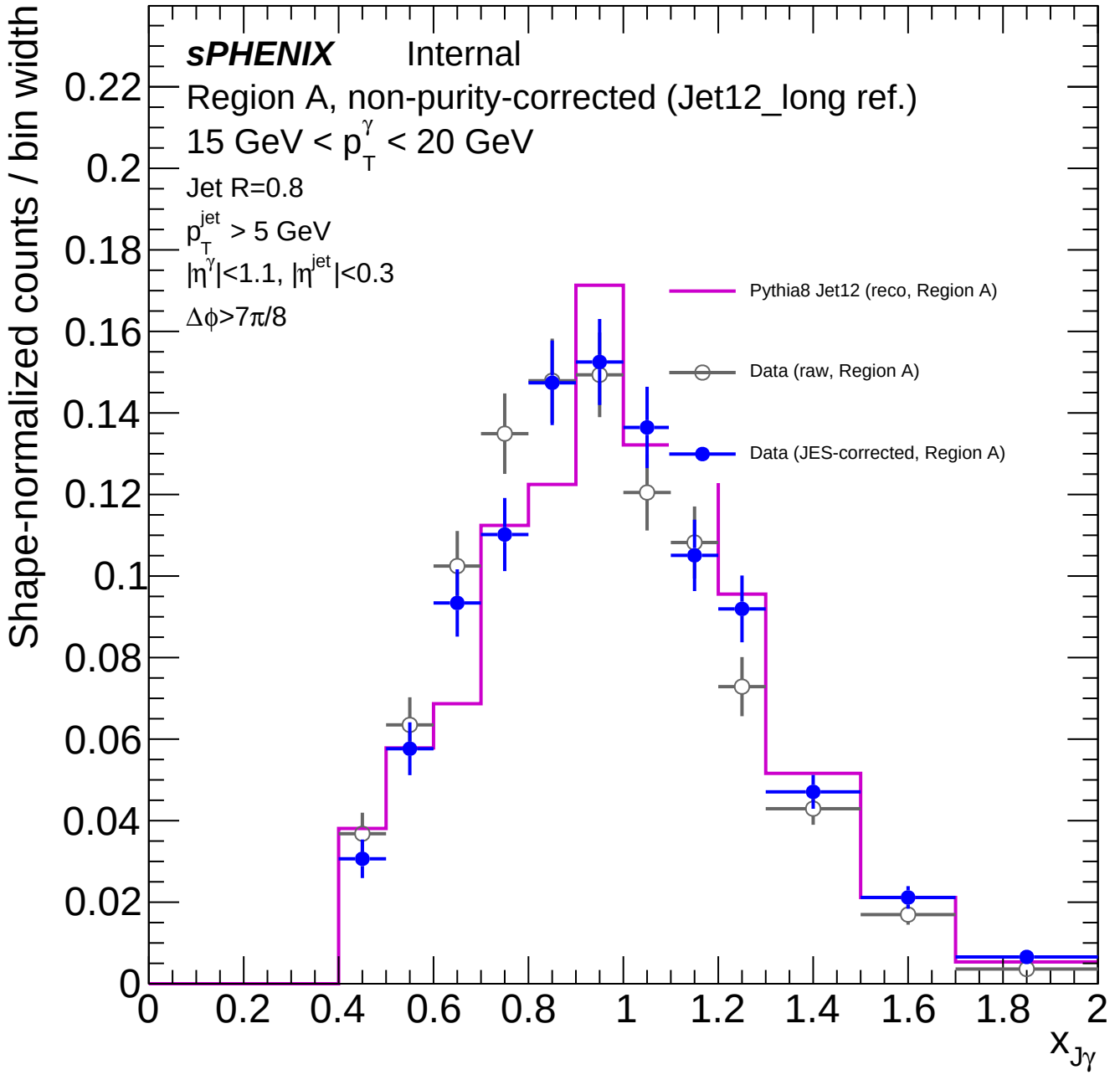

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
Jet R=0.7
|η^γ|<1.1, |η^jet|<0.4
Δϕ>7π/8
<x_Jγ>
Pythia8 Jet12 (reco, Region A)
Data (reco, Region A)
Data/MC
Raw ratio
Corrected ratio
Data to MC (Jet12) JES (shape χ²) = 0.9714 ± 0.0012
p_T^γ [GeV]

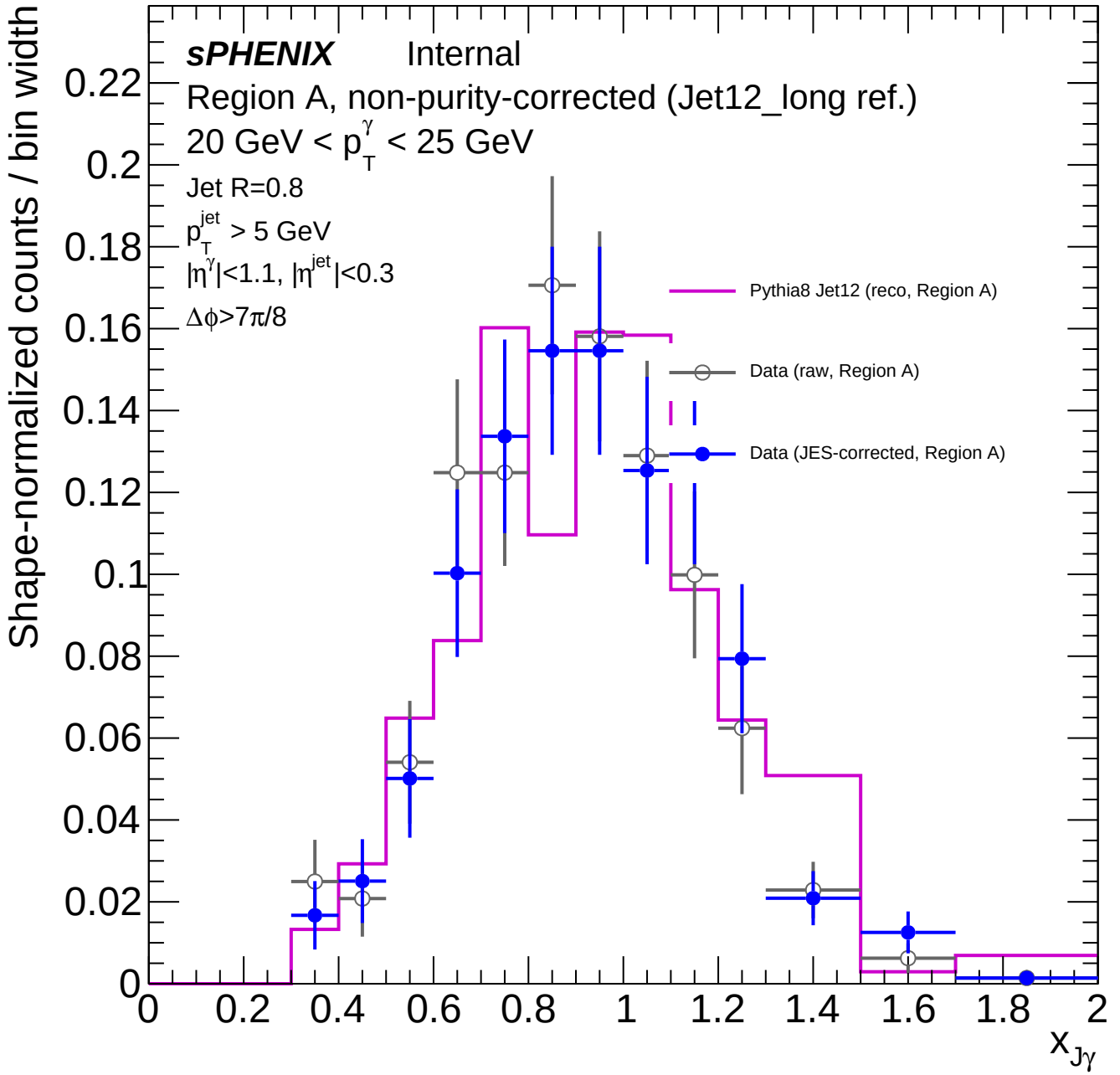

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.7
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.4
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

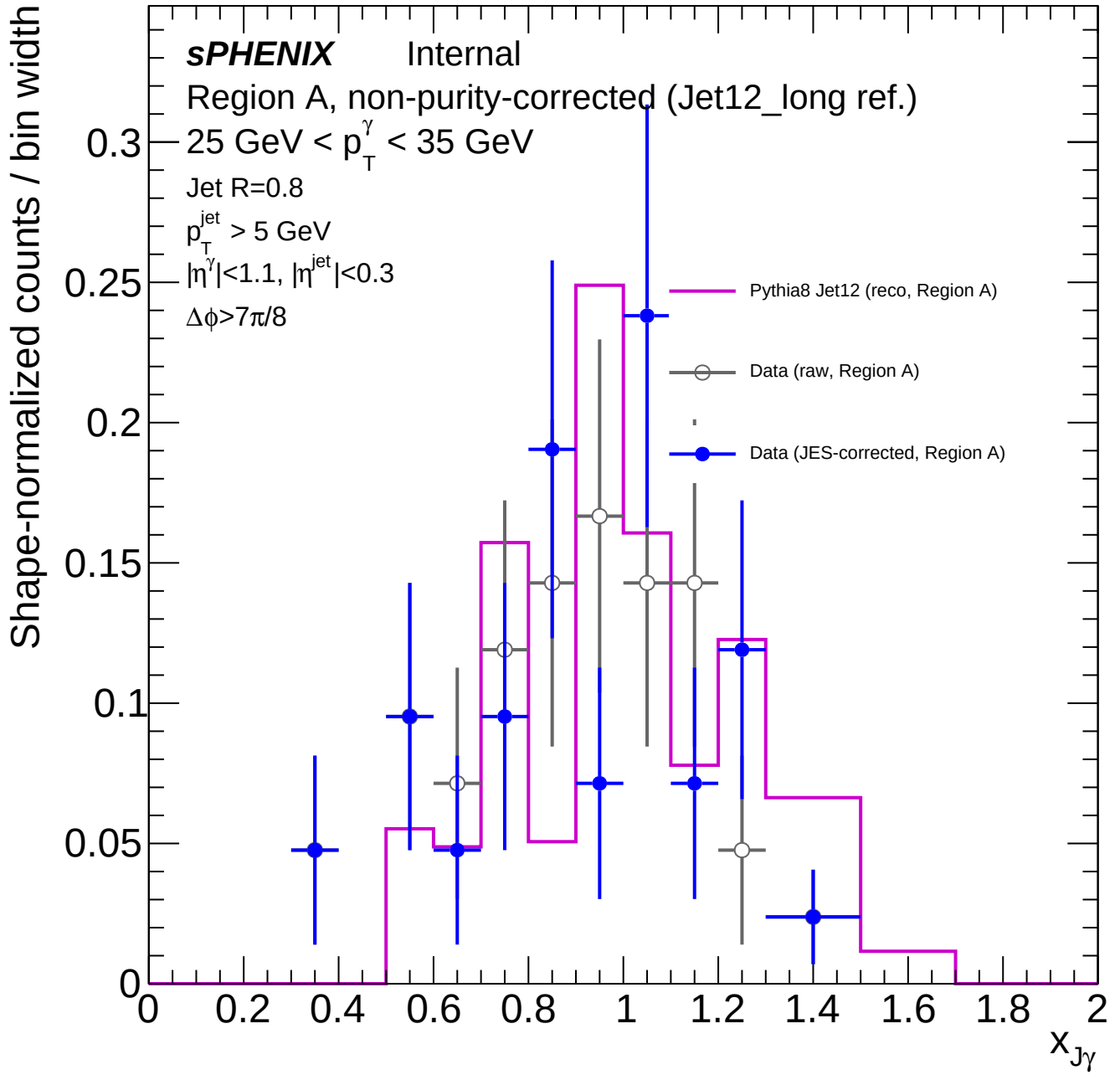

sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.7
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.4
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.7
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.4
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
Jet R=0.8
|η^γ|<1.1, |η^jet|<0.3
Δϕ>7π/8
<x_Jγ>
Pythia8 Jet12 (reco, Region A)
Data (reco, Region A)
Data/MC
Raw ratio
Corrected ratio
Data to MC (Jet12) JES (shape χ²) = 0.9661 ± 0.0008
p_T^γ [GeV]


sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
15 GeV < p_T^γ < 20 GeV
Jet R=0.8
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.3
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
20 GeV < p_T^γ < 25 GeV
Jet R=0.8
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.3
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ


sPHENIX Internal
Region A, non-purity-corrected (Jet12_long ref.)
25 GeV < p_T^γ < 35 GeV
Jet R=0.8
p_T^jet > 5 GeV
|η^γ|<1.1, |η^jet|<0.3
Δφ>7π/8
Pythia8 Jet12 (reco, Region A)
Data (raw, Region A)
Data (JES-corrected, Region A)
Shape-normalized counts / bin width
x_Jγ
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2